



UNIVERSITY OF SRI JAYEWARDENEPURA

Faculty of Applied Sciences

Department of Statistics

Exploring Dengue in Sri Lanka

A Report on an Interactive Dengue Surveillance Dashboard

Course: STA 4922 Data Visualization

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Index Number: AS2022434

BSc (Honours) in Statistics - Year 4

Dashboard URL:

<https://shenan12.github.io/dengue-health-data-dashboard/>

GitHub Repository:

<https://github.com/Shenan12/dengue-health-data-dashboard>

July 16, 2026

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Introduction

Dengue fever is a major public health challenge in Sri Lanka. Transmitted primarily by the *Aedes aegypti* mosquito, the disease requires continuous surveillance to detect outbreaks and understand regional spread. This report outlines an interactive dashboard built using Quarto and R, designed to help users explore weekly dengue surveillance data across the country through an accessible, highly visual format that facilitates both high-level summaries and granular, district-level investigations.

Data

All data utilized in this project originates directly from the `denguedatahub` R package. The primary dataset, `srilanka_weekly_data`, contains reported case counts across Sri Lanka from late 2006 onwards. To ensure consistent spatial mapping downstream, cases reported under the Kalmunai health district were explicitly merged into the Ampara district during data preparation. Additionally, the `sl_dengue_serotype` dataset was incorporated to track the specific virus strains circulating during different years.

Methods

Software and Data Preparation

The dashboard and all related analyses were developed using R. Data manipulation, date parsing, and aggregation into national and annual totals were handled utilizing the `tidyverse` suite and `lubridate`. To ensure accurate time series analysis, epidemiological week labels were generated, and the national series was converted into a `tsibble` object. Missing weeks were explicitly filled with zero cases to allow for seamless seasonal decomposition utilizing `feasts` and `fabletools`, with additional time series support from `xts`.

Interactive visualizations were constructed with `plotly` and `dygraphs`, while dynamic data tables were rendered using the `DT` package. Spatial data processing and mapping were implemented through `leaflet`, `sf`, `ceylon`, and `rmapshaper`.

Dashboard Structure and Custom Styling

The dashboard is organized into seven primary navigation tabs: Overview, National Trends, Trend and Seasonality, District Explorer, Spatial Distribution, Serotypes, and Data.

To ensure a modern, professional appearance, a `custom.css` file was integrated to override the default Quarto styling. This custom stylesheet applies clean typography using the "Inter" and "Space Grotesk" fonts. Visual hierarchy was improved by softening borders and adding subtle hover shadow effects to the `bslib-card` elements. Value boxes were specifically

styled to maintain white iconography against custom background colors, and the interactive data tables were enhanced by restyling the export buttons.

Visualisation Techniques

Several interactive techniques were employed to maximize user engagement:

- **National and District Trends:** An interactive area chart captures the national weekly timeline. A multi-line chart highlights the ten districts with the highest historical case counts using the `Paired` color palette for clear distinction.
- **District Explorer:** A dropdown menu allows users to isolate and view the timeline of any single district, toggled via `plotly's update` method. Range selector buttons and a range slider provide fine-grained control over the time window.
- **Time Series Decomposition:** An STL (Seasonal and Trend decomposition using Loess) model separated the raw national data into an underlying trend, a repeating 52-week seasonal cycle, and an irregular remainder component.
- **Spatial and Serotype Mapping:** A choropleth map was constructed over Sri Lanka using Leaflet and spatial polygons from the `ceylon` package, applying a sequential color palette to represent annual case intensity. A complementary heatmap orders districts by total cumulative cases. Serotype data was visualized using tile plots and horizontal bar charts.
- **Interactive Data Tables:** The Data tab provides raw access to the surveillance and serotype datasets using the `DT` package, equipped with filtering capabilities and buttons for CSV and Excel export.

Key Findings

Aggregate Metrics and the 2017 Epidemic

The dataset captures a massive total of 803,074 reported dengue cases over the surveillance period. The most striking visual element across the dashboard is the unprecedented 2017 epidemic, driven by severe flooding from the southwest monsoon. During Week 28 of 2017, national reported cases peaked at a staggering 10,590 in a single week. While 2017 remains the most extreme outlier, the data also reveals significant resurgences around 2020 and in the 2023 to 2024 period.

Spatial Distribution and District-Level Burden

The multi-line comparison and heatmap confirm that the Western Province bears the highest historic case burden, with Colombo and Gampaha consistently reporting the highest numbers

due to high population densities. However, severe regional spikes exist outside this province; for instance, Kandy recorded 14,204 cases during the 2017 peak, surpassing historically burdened districts like Kalutara.

The Spatial Distribution tab further validates this geographic concentration. The interactive choropleth map vividly illustrates the density of the outbreaks; selecting the year 2017 reveals that Gampaha alone recorded 28,818 annual cases, visually dominating the map with the darkest shading. This spatial perspective confirms that while dengue is endemic nationwide, transmission intensity is highly localized in densely populated urban centers.

The District Explorer provides localized context without visual clutter. Filtering to the Ampara district reveals a distinct timeline where localized surges of over 400 weekly cases bridged late 2017 and early 2018, demonstrating how regional outbreaks can sometimes trail or differ slightly from the national average.

Seasonal and Long-Term Patterns

The STL decomposition clearly isolates the cyclical behavior of the disease. The seasonal component confirms a reliable bimodal pattern every year. Cases consistently rise during approximately weeks 20 to 30, corresponding to the southwest monsoon season, followed by a smaller secondary rise later in the year during the northeast monsoon period. The remainder component of the decomposition shows a massive spike in 2017, mathematically confirming that the outbreak was highly anomalous and unpredictable based on standard seasonal or trend expectations.

Serotype Circulation

Six distinct serotype classifications have been identified in the dataset, including genetic variations and sub-lineages like DENV (IIIA) and DENV (IIIB). DENV-1, DENV-4, and DENV (IIIB) are the most consistently detected, each appearing in four separate reporting years. The frequent co-circulation of overlapping serotypes is a critical public health indicator. Prior infection with one strain does not grant immunity to the others, and secondary infections with different serotypes are associated with a significantly higher risk of severe disease.

Conclusion

This dashboard translates complex public health data into an explorable format, highlighting dengue's concentration in the Western Province and its monsoon-driven seasonal cycles. By integrating tools like the choropleth map, District Explorer, and STL decomposition, it provides a comprehensive view of predictable transmission risks alongside anomalous crises like the 2017 epidemic.

References

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